



GTC4Lusso USA - Refitting the complete PTU - Latest Update: 2021-01-26

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C4.03 Refitting the complete PTU

Refitting the PTU



Tightening torque		Nm	Class
Fastening PTU to engine	Screw	40 Nm	A
	Nut	20 Nm	A
Fastening ground connection on PTU Power Unit	Nut	9 Nm	B
	Screw	9 Nm	B
Fastening PTU hydraulic system fluid reservoir bracket to chassis	Screw	20 Nm	B
Fastening PTU hydraulic system pipes	Screw	9 Nm	B
	Nut	9 Nm	B
Fastening PTU wiring harness ground	Screw	18 Nm	B
	Screw	9 Nm	B
Fastening PTU gear oil pipes Products to use: AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK	Screw	9 Nm	B
Fastening PTU wiring harness to chassis	Screw	9 Nm	B
Fastening the PTU ECU	Screw	5 Nm	B



The workshop tests performed after installing a new component must include tests of all safety system components. If the "Safety NDA" cycle performed with the Deis tester is failed, the PTU is NOT in safe mode or the PTU registers any errors, do NOT return the vehicle to the customer.



In the event of a Pcode with the additional code TSR, contact the SAT Help Desk.



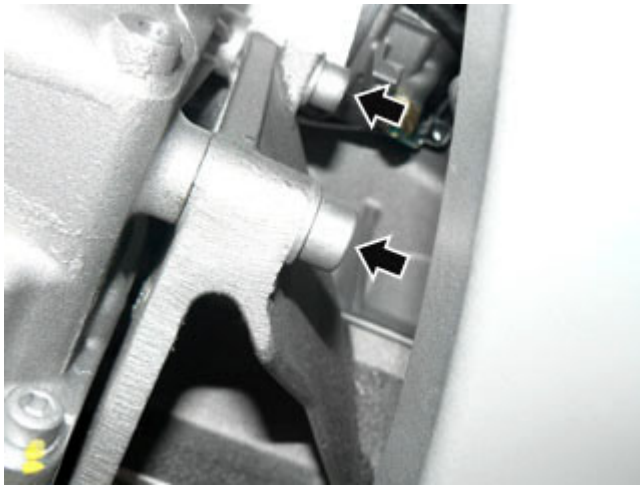
- Replace the indicated O-ring on the engine.



- Taking care not to knock any surrounding components, lift the complete PTU (D) into the engine compartment using the hydraulic jack and tool (AV 8353).



- Using the hydraulic jack and the tool, (AV 8353), mate the PTU with the engine, engaging the respective primary shaft with the decoupling shaft (C) installed on the engine.
- Check that the primary PTU shaft and the decoupling shaft (C) are perfectly aligned and mated correctly.



- Tighten the screws indicated.

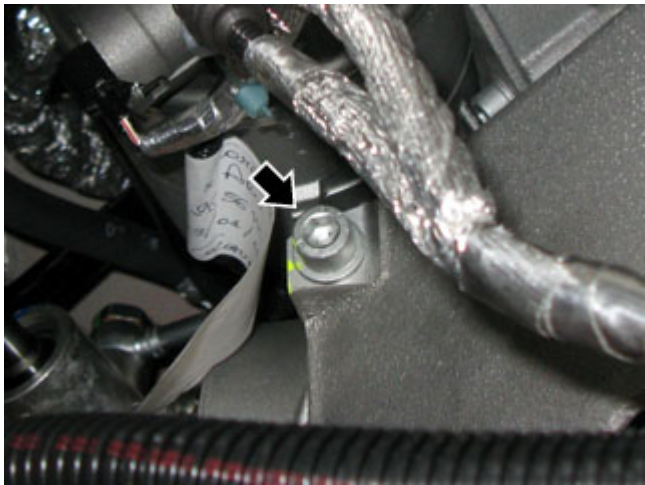


- Lower the lower pads of the tool (**AV 8353**) with the respective knobs (**13**).



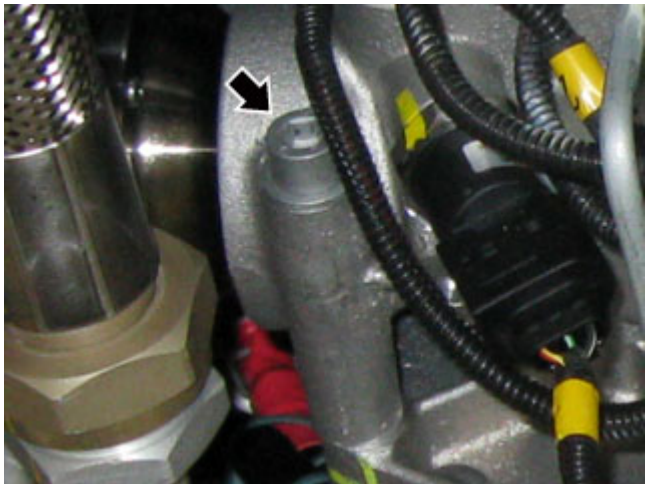
- Undo the indicated screws.
- Detach the lateral support bracket of the tool (**AV 8353**) from the front LH axle shaft flange.

- Remove the tool (**AV 8353**) from the hydraulic jack.
- Apply and hand-tighten all fasteners securing the PTU to the engine, then tighten definitively in the sequence given below.



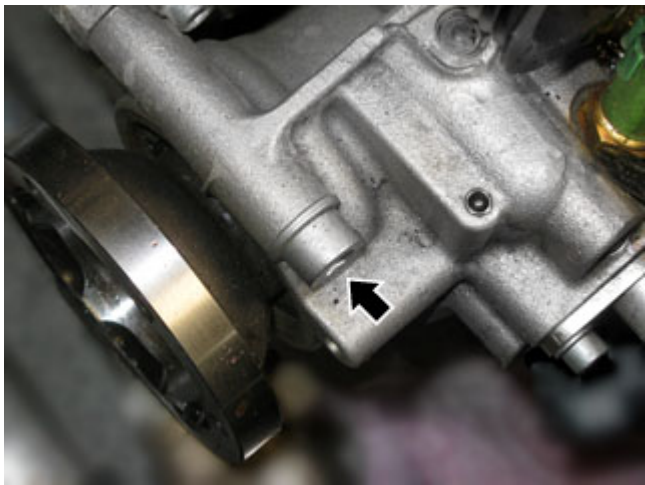
- Hand tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A



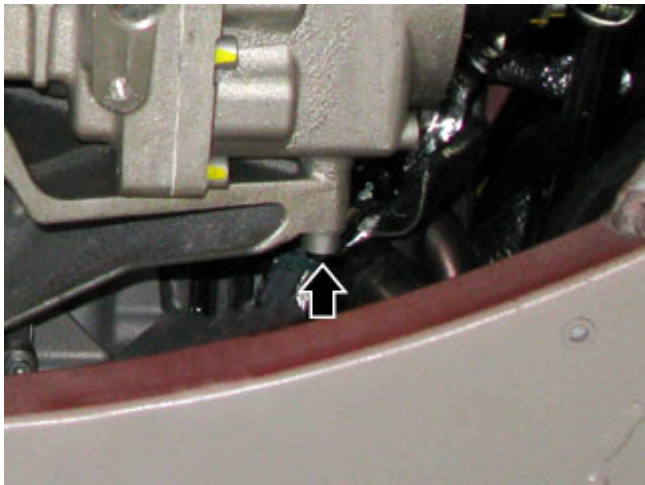
- Hand tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A



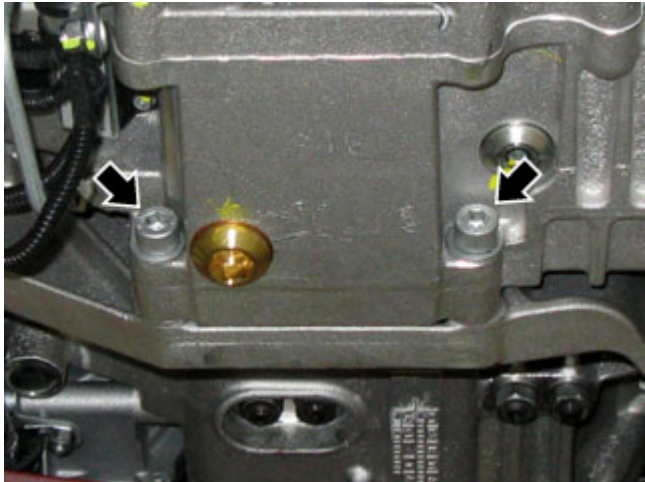
- Hand tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A



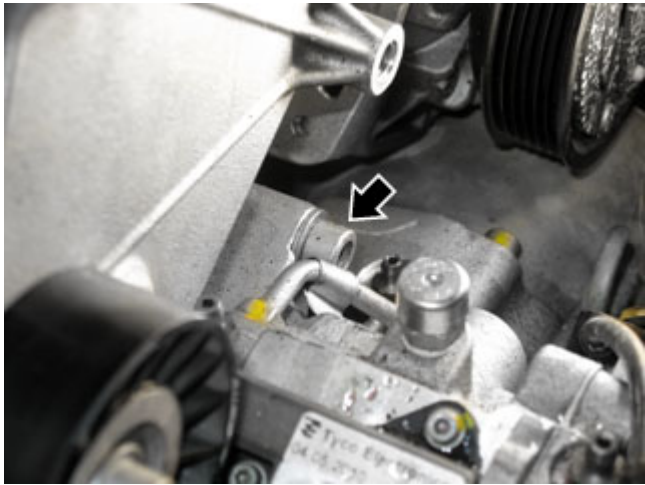
- Hand tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A



- Tighten the indicated screws.

		
Tightening torque	Nm	Class
Screw	40 Nm	A



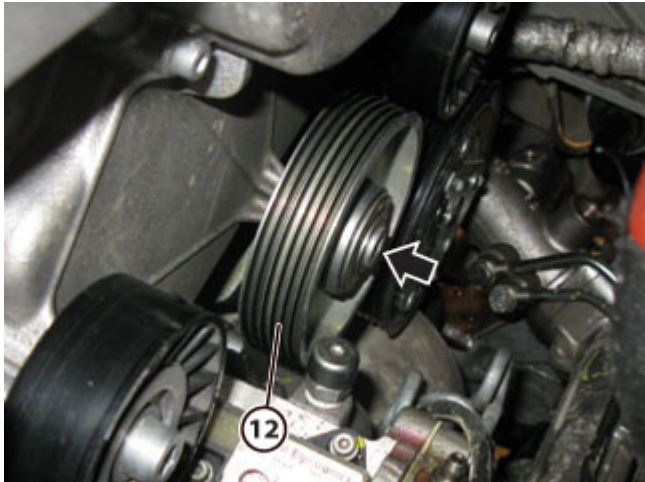
- Hand tighten the indicated screw.

		
Tightening torque	Nm	Class
Screw	40 Nm	A

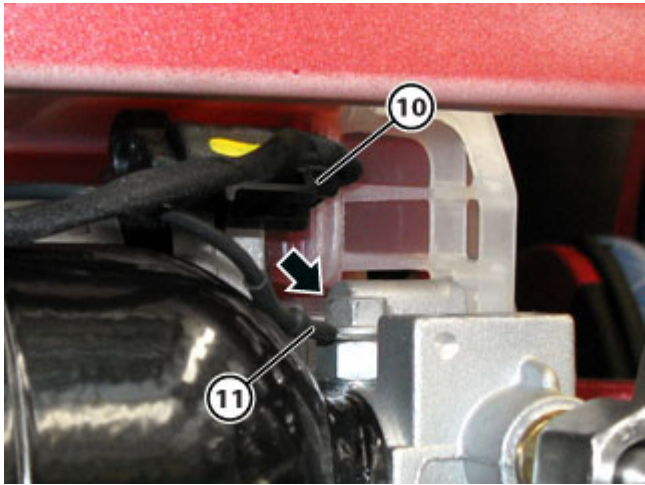


- Tighten the indicated nut.

		
Tightening torque	Nm	Class
Nut	20 Nm	A



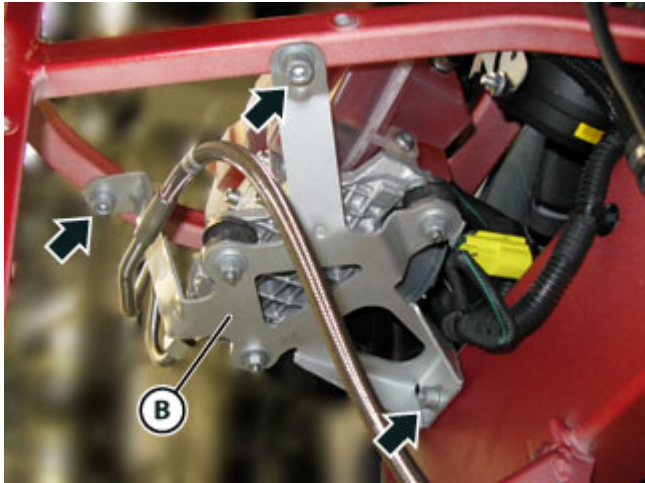
- Fit the pulley (12).
- Tighten the indicated screw.



- Connect the ground cable (11) by tightening the indicated screw.

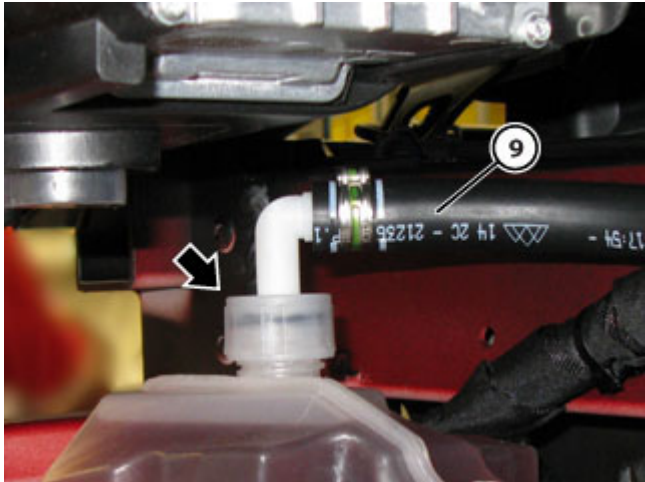
		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Connect the connector (10).

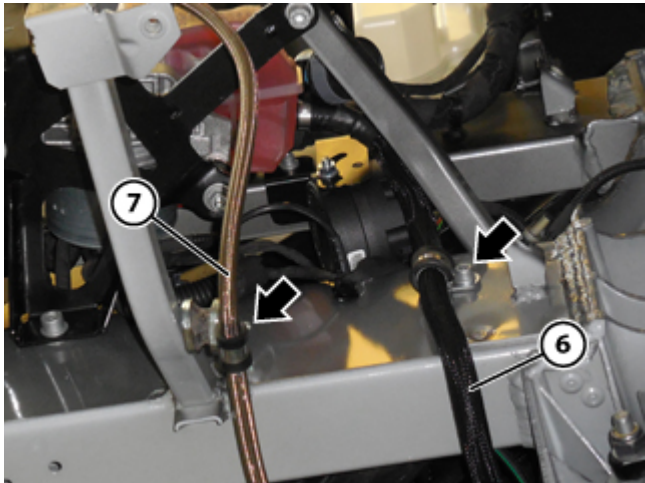


- Fit the bracket **(B)**, complete with the PTU pneumatic actuator oil reservoir.
- Tighten the indicated screws.

		
Tightening torque	Nm	Class
Screw	20 Nm	B



- Check that the seal of the threaded ring indicated is in the correct position.
- Connect the pipe **(9)**, tightening the indicated threaded ring.
- Check that the threaded ring indicated is fastened securely.

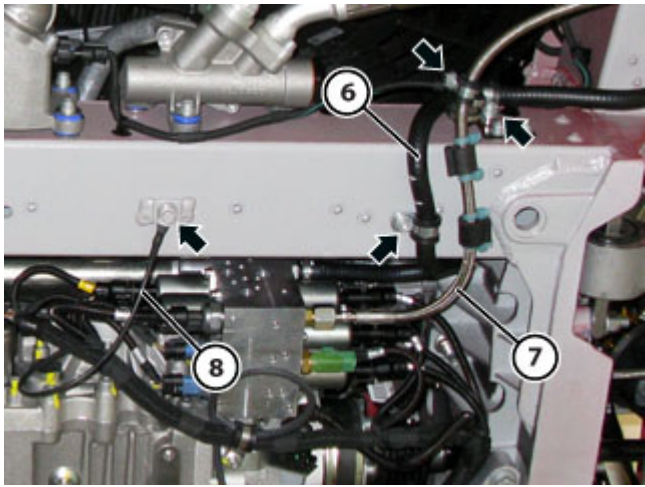


- Fasten the PTU system hydraulic actuator system pipe **(7)**, tightening the indicated nut.

		
Tightening torque	Nm	Class
Nut	9 Nm	B

- Fasten the PTU system hydraulic actuator system pipe **(6)**, tightening the indicated screw.

		
Tightening torque	Nm	Class
Screw	9 Nm	B



- Connect the ground cable (8) by tightening the indicated screw.

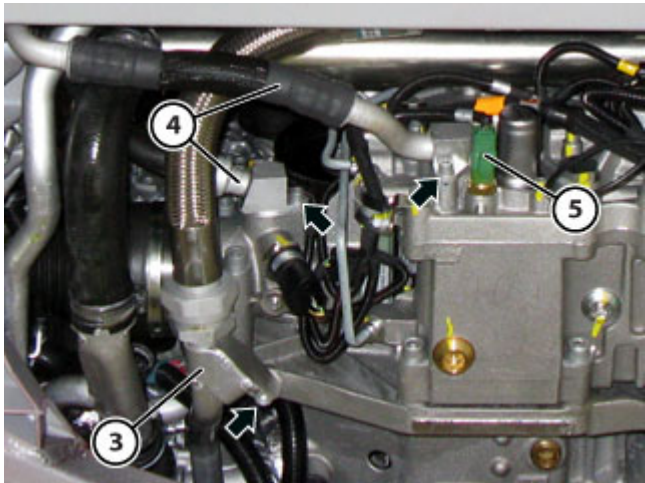
		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Fasten the PTU system hydraulic actuator system pipe (7), tightening the indicated nut.

		
Tightening torque	Nm	Class
Nut	9 Nm	B

- Fasten the PTU system hydraulic actuator system pipe (6), tightening the indicated screws.

		
Tightening torque	Nm	Class
Screw	9 Nm	B



- Connect the connector (5).
- Apply **AREXONS SYSTEM 52A43 - MEDIUM STRENGTH THREAD LOCK** to the indicated screws fastening the pipes (8).
- Connect the pipes (4), replacing the respective O-rings, and tighten the respective screws indicated.

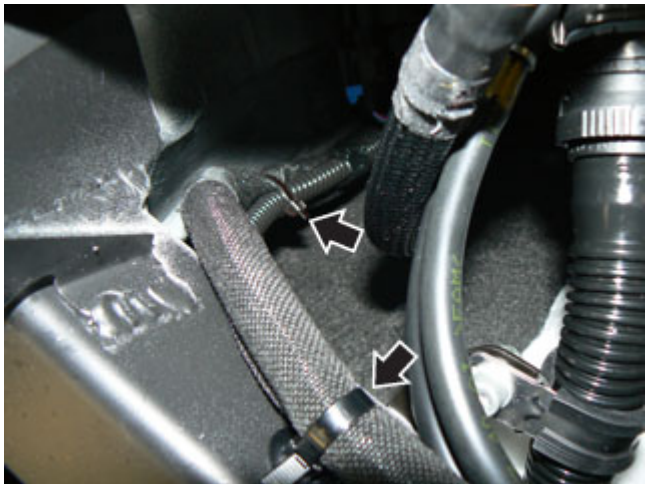
		
Tightening torque	Nm	Class
Screw	9 Nm	B

- Fit the engine oil pipe (3) and tighten the respective fastener screw indicated.

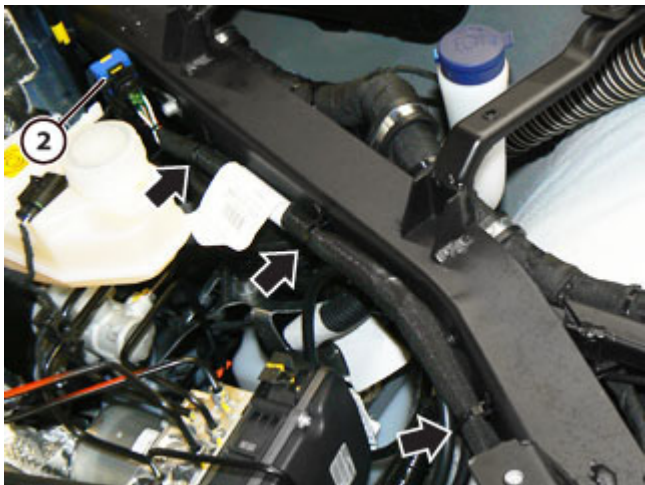


- Fasten the wiring harness by tightening the indicated screw on the rubber-lined bracket.

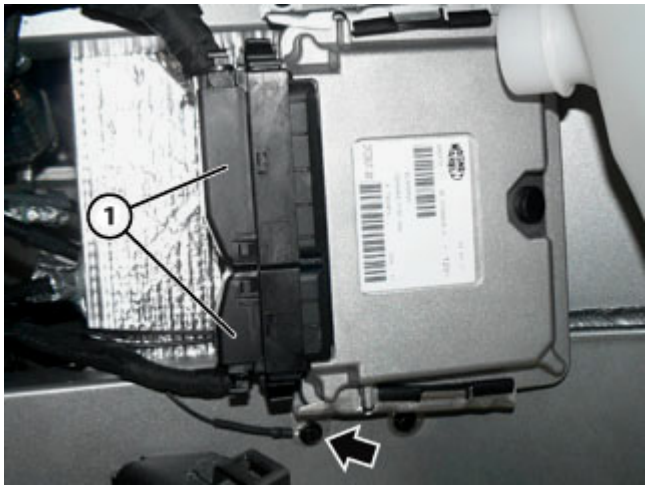
		
Tightening torque	Nm	Class
Screw	9 Nm	B



- Secure the wiring harness by fastening the indicated clips.



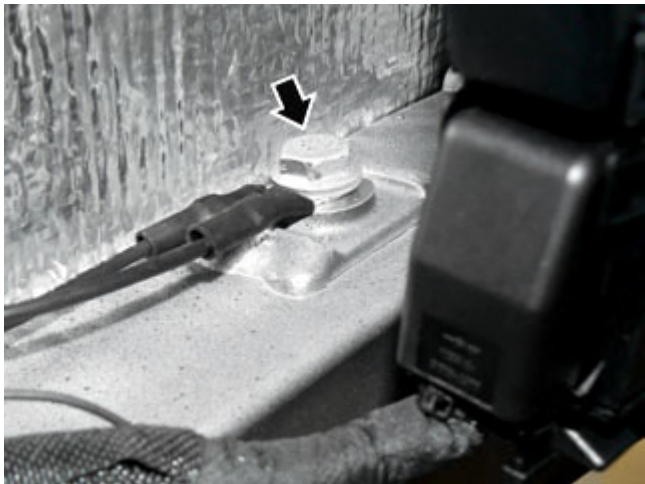
- Connect the connector (2) and secure the respective cable, fastening the indicated clips.



- Connect the ground cable, tightening the indicated screw.

		
Tightening torque	Nm	Class
Screw	5 Nm	B

- Connect the connectors (1).



- Connect the ground cables, tightening the indicated screw.

		
Tightening torque	Nm	Class
Screw	18 Nm	B

- Refit the front wheelhouses ( E3.05).
- Refit the front axle shafts ( C5.01).
- Refit the front stabiliser bar ( D4.05).
- Refit the steering box ( D5.04).
- Refit the front bumper ( E3.03).
- Refit the engine compartment cross-member ( E2.04).
- Connect the battery ( F2.01).
- Fill the PTU gear oil system ( C4.06).
- Connect the DEIS tester to the diagnostic socket ( F2.09).
- Perform the cycle "20_NDA Gearbox Self Learning" with the DEIS diagnostic tester.